

Internship Final Report

Student Name: Chinmayi S

University: VTU

Major: Loan Prediction using Random Forest

Internship Duration: September 1st, 2025 - September 30th, 2025

Company: ShadowFox

Domain: AI-ML

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

Objectives

- Train a **Random Forest classifier** on a loan prediction dataset.
- Interpret model predictions using **SHAP (SHapley Additive exPlanations)** to understand feature contributions.
- Generate visual reports (waterfall plots, force plots, and summary plots) for model explainability.
- Document the analysis in a structured report for internship submission.

Tasks and Responsibilities

- **Preprocessed the dataset and created a pipeline for the Random Forest model.**

Trained and evaluated the classifier, obtaining the final test accuracy.

Generated SHAP explanations for individual predictions and global feature importance. Saved visualizations as images and

compiled a PDF report for all samples.

Prepared a GitHub repository with README, notebook, and supporting files. Recorded a proof-of-work video for LinkedIn showcasing the project workflow.

Learning Outcomes

Gained practical experience with **model interpretability** using SHAP. Learned to handle **multi-output SHAP explanations** and generate visual plots.

- Developed skills in **Python automation** to save plots and reports programmatically.
- Improved **documentation skills**, including preparing GitHub repos, README, and PDF reports.

Challenges & Solutions

- **Challenge:** SHAP force and waterfall plots behave differently in v0.20+.
Solution: Used **shap.Explanation objects** and extracted class-specific SHAP values.
- **Challenge:** Saving interactive force plots as static images.
Solution: Used matplotlib=True in shap.force_plot and saved with plt.savefig().
- **Challenge:** Combining multiple images into a single PDF.
Solution: Used **PIL.Image** to resize and merge images before saving.
- **Challenge:** Feature alignment issues in preprocessed data.
Solution: Ensured all required columns from training data were present in new samples.

Conclusion

- **Successfully trained and evaluated a Random Forest model for loan prediction.**
- **Generated interpretable SHAP visualizations for both individual and global predictions.**
- **Produced a comprehensive PDF report and GitHub repository for documentation.**
- **Prepared all deliverables required for internship proof-of-work.**

Acknowledgments

- **Grateful to ShadowFox for providing the AI/ML internship opportunity.**
- **Thank you to mentors and peers for guidance and support throughout the project.**
- **Appreciation to online resources and documentation that helped with SHAP explainability techniques.**